TECHNICAL DILIGENCE REVIEW - RECORD 011

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-011

1. Engagement scope, assessors and evidence conventions

The present report was prepared by Brandywine Technology Partners for Ardent Peak and covers the security architecture, penetration-test findings and vulnerability management of the platform at Helioscope Robotics, Inc. as at 2026-01-23. 2026-01-23 is the evidence cut-off for the report at TDR-2025-38, whose subject is the security architecture, penetration-test findings and vulnerability management of the platform. K. Vandermeer was granted read access to the code, the pipeline and the incident tracker for the purposes of report TDR-2025-38; production access was not requested. K. Vandermeer led the work behind the report at TDR-2025-38, combining document review, 11 interviews and hands-on inspection of running systems. Any figure in this report that the reviewer could not trace to a system is attributed to the person who gave it.

In this report the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Work on the present report ran over 5 weeks and included 11 interviews and read-only access to the engineering repositories. Whatever weight a reader gives the individual findings, TDR-2025-38 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Taking the sections above together, there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Readers comparing the present report with an earlier technical report should note that the service inventory was renamed in between.

TDR-2025-38 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. In this report the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. Drawing the work together, the report at TDR-2025-38 finds that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer's conclusion in TDR-2025-38 is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Cross-references in TDR-2025-38 point to the evidence index rather than to a summary of the evidence.

The present report covers 14 of 34 services; the untested remainder is listed so a reader can see what was not reached. K. Vandermeer chose the sample for the report at TDR-2025-38 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. The judgement the report at TDR-2025-38 reaches is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer would summarise the present report in one sentence: there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer notes that the company answered every document request, though two answers arrived after the draft.

Two document requests remained partly unsatisfied when the present report was delivered, and the affected conclusions are marked. The report at TDR-2025-38 is limited by log retention: the telemetry window is ninety days, so events before 2026-01-05 could not be examined directly. Brandywine Technology Partners was instructed by Ardent Peak to examine the security architecture, penetration-test findings and vulnerability management of the platform; the resulting report is TDR-2025-38, dated 2026-01-23. The subject of the present report is the security architecture, penetration-test findings and vulnerability management of the platform, examined for Ardent Peak by Brandywine Technology Partners over 5 weeks ending 2026-01-23. The company reviewed a draft of this report for factual accuracy and its corrections are reflected without altering any conclusion.

Interviewees for the present report were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. 8 follow-up items arose from the interview programme for the present report; 2 were still open at delivery. The overall conclusion of this report is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. On the evidence available by 2026-01-23, the report at TDR-2025-38 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer has stated the limitations at the point they bite rather than collecting them into a disclaimer.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-38)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-11-001	Site Provisioning Service: configuration export	configuration export	2026-08-27	the sensor vendor	established provenance
EV-11-002	Map Authoring Studio: runbook	runbook	2026-10-04	Helioscope field reliability	established provenance
EV-11-003	Analytics Warehouse Loader: configuration export	configuration export	2026-09-12	Helioscope engineering	verified a stated objective
EV-11-004	Fleet Allocator: design record	design record	2026-05-22	Helioscope engineering	evidenced a remediation
EV-11-005	Localization Engine: policy document	policy document	2026-01-17	Helioscope engineering	evidenced a remediation
EV-11-006	Safety Controller: audit output	audit output	2025-11-23	the safety controller vendor	evidenced a remediation
EV-11-007	Hardware-in-the-Loop Rig: test report	test report	2025-06-22	Meridian Assurance	evidenced an approval
EV-11-008	Hardware-in-the-Loop Rig: runbook	runbook	2026-05-16	Helioscope field reliability	verified a stated objective
EV-11-009	hardware-in-the-loop, bench: runbook	runbook	2025-12-22	Helioscope engineering	established a dependency
EV-11-010	WMS Connector: design record	design record	2026-10-08	Helioscope safety	evidenced a remediation
EV-11-011	Robot Shadow Store: interview minute	interview minute	2026-07-04	the safety controller vendor	established provenance
EV-11-012	Billing Usage Meter: interview minute	interview minute	2026-04-18	Helioscope security	corroborated an incident narrative
EV-11-013	release packaging and signing: architecture note	architecture note	2026-03-02	the safety controller vendor	evidenced a remediation
EV-11-014	Gripper Controller: vendor statement	vendor statement	2026-01-20	Helioscope security	evidenced a remediation
EV-11-015	performance smoke, cloud: runbook	runbook	2025-05-02	Helioscope safety	corroborated an incident narrative
EV-11-016	Map Authoring Studio: release record	release record	2025-06-18	Redwater Security	established provenance
EV-11-017	contract tests, warehouse connectors: policy document	policy document	2025-06-30	Helioscope field reliability	supported a capacity conclusion
EV-11-018	unit tests, cloud services: incident post-mortem	incident post-mortem	2025-06-08	Helioscope security	supported a capacity conclusion
EV-11-019	Billing Usage Meter: configuration export	configuration export	2026-03-30	the safety controller vendor	established a dependency
EV-11-020	Speed and Separation Monitor: policy document	policy document	2025-08-07	Helioscope security	evidenced a remediation
EV-11-021	Order Broker: measurement extract	measurement extract	2025-06-24	Helioscope field reliability	established provenance
EV-11-022	Safety Controller: runbook	runbook	2026-02-01	the safety controller vendor	verified a stated objective
EV-11-023	Fleet Allocator: policy document	policy document	2026-08-12	the sensor vendor	supported a coverage figure
EV-11-024	unit tests, robot software: runbook	runbook	2026-05-02	Helioscope security	established a dependency
EV-11-025	Hardware-in-the-Loop Rig: vendor statement	vendor statement	2026-04-13	Meridian Assurance	corroborated an incident narrative
EV-11-026	Fleet Health Monitor: design record	design record	2026-05-27	Helioscope field reliability	established provenance
EV-11-027	Edge Gateway Agent: runbook	runbook	2026-09-07	Redwater Security	evidenced an approval
EV-11-028	Wheel Odometry Filter: measurement extract	measurement extract	2025-09-08	the safety controller vendor	corroborated an incident narrative
EV-11-029	release packaging and signing: vendor statement	vendor statement	2026-03-26	Helioscope field reliability	established a dependency

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-11-030	Identity and Authorization: interview minute	interview minute	2026-08-24	the safety controller vendor	established provenance
EV-11-031	Hardware-in-the-Loop Rig: interview minute	interview minute	2025-07-05	Redwater Security	evidenced a remediation
EV-11-032	dependency and licence scan: design record	design record	2025-12-12	Redwater Security	corroborated an incident narrative
EV-11-033	WMS Connector: policy document	policy document	2026-06-18	Meridian Assurance	established provenance
EV-11-034	E-Stop Chain: measurement extract	measurement extract	2026-03-16	Helioscope safety	established provenance
EV-11-035	contract tests, warehouse connectors: vendor statement	vendor statement	2026-07-18	the safety controller vendor	supported a coverage figure
EV-11-036	Order Broker: audit output	audit output	2026-10-11	Redwater Security	supported a capacity conclusion

1. Engagement scope, assessors and evidence conventions - part 2 of 2

The 1. Engagement scope, assessors and evidence conventions begun above continues from this point, and its row references run consecutively across the parts rather than restarting. The break is editorial.

Because the basis of the 1. Engagement scope, assessors and evidence conventions is unchanged, figures in this part may be compared directly with those already given.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-38) (continued, part 2 of 2)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-11-037	Map Authoring Studio: release record	release record	2025-07-11	the safety controller vendor	established provenance
EV-11-038	Remote Operations Console: design record	design record	2025-05-29	Helioscope engineering	established provenance
EV-11-039	Traffic Manager: design record	design record	2026-04-28	Redwater Security	established provenance
EV-11-040	Charge Scheduler: vendor statement	vendor statement	2026-03-14	Meridian Assurance	evidenced a remediation
EV-11-041	Video Snippet Service: interview minute	interview minute	2026-08-14	Redwater Security	evidenced an approval
EV-11-042	Hardware-in-the-Loop Rig: policy document	policy document	2026-02-25	Meridian Assurance	corroborated an incident narrative
EV-11-043	Video Snippet Service: architecture note	architecture note	2026-03-14	the safety controller vendor	established provenance
EV-11-044	contract tests, warehouse connectors: configuration export	configuration export	2026-05-10	Helioscope field reliability	established a dependency
EV-11-045	Speed and Separation Monitor: vendor statement	vendor statement	2025-12-17	Helioscope field reliability	established provenance
EV-11-046	E-Stop Chain: architecture note	architecture note	2026-04-05	the sensor vendor	supported a coverage figure
EV-11-047	Payload Scale Interface: policy document	policy document	2026-04-30	Helioscope security	verified a stated objective
EV-11-048	Config Distribution: architecture note	architecture note	2026-08-03	the safety controller vendor	evidenced an approval
EV-11-049	hardware-in-the-loop, bench: architecture note	architecture note	2025-07-24	the safety controller vendor	established a dependency
EV-11-050	Battery Management Service: test report	test report	2025-12-18	Helioscope safety	supported a capacity conclusion

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-11-051	OTA Updater: architecture note	architecture note	2025-10-25	Helioscope security	corroborated an incident narrative
EV-11-052	Gripper Controller: policy document	policy document	2025-06-09	Meridian Assurance	established provenance
EV-11-053	Analytics Warehouse Loader: release record	release record	2025-12-21	the safety controller vendor	corroborated an incident narrative
EV-11-054	dependency and licence scan: architecture note	architecture note	2025-08-21	Redwater Security	corroborated an incident narrative
EV-11-055	simulation suite, nightly: release record	release record	2026-02-24	Meridian Assurance	evidenced a remediation
EV-11-056	release packaging and signing: runbook	runbook	2025-12-01	Helioscope field reliability	corroborated an incident narrative
EV-11-057	Charge Scheduler: test report	test report	2025-11-01	Meridian Assurance	evidenced a remediation
EV-11-058	Event Bus: policy document	policy document	2026-07-10	Meridian Assurance	established provenance
EV-11-059	container image vulnerability scan: architecture note	architecture note	2026-05-21	Helioscope field reliability	established a dependency
EV-11-060	Speed and Separation Monitor: test report	test report	2025-05-27	Helioscope field reliability	verified a stated objective
EV-11-061	build, robot firmware images: policy document	policy document	2025-11-25	Meridian Assurance	supported a capacity conclusion
EV-11-062	Remote Operations Console: test report	test report	2025-07-09	Meridian Assurance	evidenced an approval
EV-11-063	Telemetry Ingest: interview minute	interview minute	2026-05-16	the safety controller vendor	verified a stated objective
EV-11-064	Traffic Manager: policy document	policy document	2026-02-08	Redwater Security	evidenced a remediation
EV-11-065	Telemetry Ingest: runbook	runbook	2025-11-24	Redwater Security	established provenance
EV-11-066	unit tests, robot software: interview minute	interview minute	2026-08-11	Meridian Assurance	supported a capacity conclusion
EV-11-067	Battery Management Service: interview minute	interview minute	2026-10-19	Helioscope engineering	supported a capacity conclusion
EV-11-068	contract tests, warehouse connectors: design record	design record	2025-12-24	Redwater Security	verified a stated objective
EV-11-069	dependency and licence scan: design record	design record	2026-06-21	Helioscope safety	evidenced a remediation
EV-11-070	Video Snippet Service: runbook	runbook	2026-05-28	the sensor vendor	evidenced an approval
EV-11-071	Map Authoring Studio: design record	design record	2026-03-17	Redwater Security	corroborated an incident narrative
EV-11-072	Payload Scale Interface: audit output	audit output	2026-08-02	the safety controller vendor	corroborated an incident narrative

2. SEC-2025-01 and the high-severity findings

Reading SEC-2025-01 against Identity and Authorization, the defect is that service tokens are issued with a twelve-hour lifetime; the discovery date is 2025-09-16. Assessment finding SEC-2025-01 produces this judgement: the underlying pattern appears in two other components that the tester did not reach. In SEC-2025-01 the exploitation path means a captured token remains usable long after the session that produced it. The evidence for assessment finding SEC-2025-01 is moderate: reproduced by the tester and confirmed in the token audit. Identity and Authorization finding SEC-2025-01 is the clearest instance in TDR-2025-38 of engineering practice being better than the documentation suggests.

Assessment finding SEC-2026-04 is assessed as follows: the risk acceptance lacks an expiry date, which is a process finding in its own right. Customer commercial data is retained in operational logs is the exposure WMS Connector finding SEC-2026-04 describes. After the fix, finding SEC-2026-04 carries residual risk described as low; payload logging requires an explicit per-tenant approval. Moderate evidence supports SEC-2026-04 against WMS Connector, namely that payloads recovered from a staging log store. SEC-2026-04 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Finding SEC-2026-12 is rated Low with a numeric band of 3.5 on the tester's scoring scale. Partially remediated is the status of Low-severity finding SEC-2026-12, dated pending. The reviewer's position on assessment finding SEC-2026-12 is that management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Evidence behind SEC-2026-12 is that keys recovered from a returned module, which the reviewer treats as strong. Low-severity finding SEC-2026-12 is reported at the evidence strength available and has not been promoted to fit the narrative.

At 3.2 the score on assessment finding SEC-2026-20 places it in the Low group. 2026-04-24 is the discovery date of finding SEC-2026-20, which finds that payload scale calibration record can be edited without an audit entry. SEC-2026-20 against Payload Scale Interface reduces to a residual risk of low; management accepted pending the calibration service redesign. Not planned is the closure position on SEC-2026-20 against Payload Scale Interface, which the ledger records as accepted. assessment finding SEC-2026-20 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Substantiating finding SEC-2025-10 is the fact that endpoint reachable and content captured, graded strong. Residual risk after remediation of SEC-2025-10 against Charge Scheduler is low. SEC-2025-10 against Charge Scheduler stands remediated; 2025-10-09 is what the record gives for closure. 2025-09-24 is the discovery date of Charge Scheduler finding SEC-2025-10, which finds that charge scheduler exposed an unauthenticated internal health endpoint with site names. Charge Scheduler finding SEC-2025-10 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

SEC-2026-02 rests on the observation that the tester read another tenant's shadow state from a test workload; the reviewer holds that to be strong. Finding SEC-2026-02 sits against Robot Shadow Store and records that robot shadow store cache lacked authorization on the internal read path. SEC-2026-02 matters because any cluster workload could read fleet position data for all tenants. SEC-2026-02 against Robot Shadow Store was graded High, band 7.6, by Meridian Assurance engagement MA-2026-1. finding SEC-2026-02 was one of two items where TDR-2025-38 could not obtain the underlying data within the engagement.

Reading SEC-2026-06 against Simulation Harness, the defect is that simulation harness cluster used a long-lived cloud role with broad read access; the discovery date is 2026-04-17. For Medium-severity finding SEC-2026-06 the reviewer notes that the underlying pattern appears in two other components that the tester did not reach. Assessment finding SEC-2026-06 stands partially remediated; pending is what the record gives for closure. SEC-2026-06 reduces to a residual risk of medium; the role was scoped down but still reads one production bucket. SEC-2026-06 against Simulation Harness is reported at the evidence strength available and has not been promoted to fit the narrative.

Fleet Health Monitor finding SEC-2026-09 translates into the following exposure: a single query can exhaust the service's memory budget. 2026-05-08 is the closure position on Low-severity finding SEC-2026-09, which the ledger records as remediated. For SEC-2026-09 against Fleet Health Monitor the reviewer notes that the remediation is adequate and the time to remediate is not. The evidence for finding SEC-2026-09 is strong: service restart induced by the tester's query. Low-severity finding SEC-2026-09 was cross-checked against the engineering record supplied for TDR-2025-38, and the two agree.

The reviewer declines to soften assessment finding SEC-2026-10: management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. On Billing Usage Meter finding SEC-2026-10 the tester's evidence was that bucket configuration read from the cloud export, and the reviewer grades it limited, resting on a configuration read rather than on a demonstrated exploitation. 2026-06-05 is the closure position on finding SEC-2026-10, which the ledger records as remediated. Low-severity finding SEC-2026-10 is rated Low with a numeric band of 4.0 on the tester's scoring scale. finding SEC-2026-10 carries a follow-up request that was satisfied after the draft of TDR-2025-38 was circulated.

Weighing SEC-2025-12 against the compensating controls, the remediation is adequate and the time to remediate is not. Reading Analytics Warehouse Loader finding SEC-2025-12, the defect is that analytics warehouse service account held write access to raw telemetry; the discovery date is 2025-09-25. Finding SEC-2025-12 arrives as a Low finding with a 4.2 score, and the reviewer did not adjust either. Residual risk after remediation of Analytics Warehouse Loader finding SEC-2025-12 is medium; management accepted the risk with a compensating audit trail. finding SEC-2025-12 carries a follow-up request that was satisfied after the draft of TDR-2025-38 was circulated.

Schedule K - Penetration-test findings by severity with residual risk (TDR-2025-38)

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-01	High	7.4	Identity and Authorization	2025-09-16	2025-11-14	low once lifetimes are cut to one hour with refresh
SEC-2025-02	High	8.1	Release Orchestrator	2025-09-17	2025-09-19	low; redaction is enforced in the logging library
SEC-2025-03	Medium	6.2	Remote Operations Console	2025-09-18	2026-01-22	low; pinning is applied at the gateway and the agent
SEC-2025-04	Medium	5.4	Map Authoring Studio	2025-09-18	2025-11-27	medium; the shortened publication path is still outstanding
SEC-2025-05	High	7.8	Telemetry Ingest	2025-09-19	2026-02-06	low; attestation is now required on the ingest path
SEC-2025-06	Medium	6.5	Video Snippet Service	2025-09-19	2025-12-18	low; the index is scoped per tenant and the identifiers are opaque
SEC-2025-07	Medium	5.9	Site Provisioning Service	2025-09-22	2026-01-16	medium; two forked templates were not re-issued until May 2026
SEC-2025-08	Medium	6.0	Event Bus	2025-09-23	2025-10-31	low; the interface is restricted to the operations subnet
SEC-2025-09	Medium	6.3	OTA Updater	2025-09-24	pending	medium; a two-person rule is designed but not enforced in the tool
SEC-2025-10	Low	3.7	Charge Scheduler	2025-09-24	2025-10-09	low
SEC-2025-11	Low	3.1	Remote Operations Console	2025-09-25	2025-10-30	low
SEC-2025-12	Low	4.2	Analytics Warehouse Loader	2025-09-25	not planned	medium; management accepted the risk with a compensating audit trail
SEC-2026-01	High	7.1	Release Orchestrator	2026-04-14	committed 2026-08-31	high until the second-approver control lands
SEC-2026-02	High	7.6	Robot Shadow Store	2026-04-15	2026-05-29	low; authorization is checked on both the cache and durable paths
SEC-2026-03	High	8.4	Config Distribution	2026-04-15	2026-06-12	low; the flag was removed and the agent refuses unsigned bundles
SEC-2026-04	Medium	5.6	WMS Connector	2026-04-16	2026-05-20	low; payload logging requires an explicit per-tenant approval
SEC-2026-05	Medium	5.2	Alarm Router	2026-04-16	committed 2026-09-30	medium; per-region credentials

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
						are scheduled but not issued
SEC-2026-06	Medium	6.1	Simulation Harness	2026-04-17	pending	medium; the role was scoped down but still reads one production bucket
SEC-2026-07	Medium	6.7	Hardware-in-the-Loop Rig	2026-04-17	committed 2026-10-31	medium; network segmentation is the only current control
SEC-2026-08	Medium	6.4	Identity and Authorization	2026-04-18	2026-06-30	low; role grants now require a second approver and are alerted
SEC-2026-09	Low	3.9	Fleet Health Monitor	2026-04-18	2026-05-08	low
SEC-2026-10	Low	4.0	Billing Usage Meter	2026-04-19	2026-06-05	low; object lock is applied and verified nightly
SEC-2026-11	Low	4.4	Gripper Controller	2026-04-19	committed 2026-11-30	medium; the safety controller limit is the compensating control
SEC-2026-12	Low	3.5	Edge Gateway Agent	2026-04-20	pending	medium; a wipe procedure exists but was not applied retroactively
SEC-2026-13	Low	3.3	Order Broker	2026-04-20	2026-05-15	low
SEC-2026-14	Low	3.8	Traffic Manager	2026-04-21	2026-06-18	low
SEC-2026-15	Low	3.0	Site Provisioning Service	2026-04-21	not planned	low; management judged the operational cost of enforcement higher than the risk
SEC-2026-16	Medium	5.8	Costmap Fusion	2026-04-22	2026-06-26	low; the socket binds to the loopback interface only
SEC-2026-17	High	7.9	Motion Planner	2026-04-22	2026-06-19	low; the port requires a signed local credential
SEC-2026-18	Low	3.4	Battery Management Service	2026-04-23	committed 2026-12-15	low; no fleet impact demonstrated
SEC-2026-19	Medium	5.1	Localization Engine	2026-04-23	committed 2026-10-15	medium; disk encryption is planned for the next compute revision
SEC-2026-20	Low	3.2	Payload Scale Interface	2026-04-24	not planned	low; management accepted pending the calibration service redesign

3. Medium and low findings, ageing and residual risk

For Map Authoring Studio finding SEC-2025-04 the reviewer notes that management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. For Map Authoring Studio finding SEC-2025-04 the record shows that reproduced, and independently observed in a field incident, which is limited, resting on a configuration read rather than on a demonstrated exploitation evidence and is graded separately from severity. At 5.4 the score on Map Authoring Studio finding SEC-2025-04 places it in the Medium group. 2025-11-27 is the closure position on finding SEC-2025-04, which the ledger records as remediated. SEC-2025-04 against Map Authoring Studio was excluded from an earlier draft of TDR-2025-38 and reinstated after the reviewer obtained the supporting record.

Remediation status on Medium-severity finding SEC-2025-08 is remediated; the record shows 2025-10-31. SEC-2025-08 against Event Bus was graded Medium, band 6.0, by Redwater Security engagement RW-2025-3. Finding SEC-2025-08 matters because a compromised application container could reconfigure topics. Moderate evidence supports Medium-severity finding SEC-2025-08, namely that reachability proven by the tester from a deployed container. assessment finding SEC-2025-08 was reviewed against the position twelve months earlier, and nothing measurable has changed.

The residual position on finding SEC-2025-02 is low; redaction is enforced in the logging library, stated after mitigation rather than assuming full success. Evidence behind SEC-2025-02 against Release Orchestrator is that the tester extracted a working token from a retained log, which the reviewer treats as moderate. Build logs printed the full authorization header is the substance of SEC-2025-02 against Release Orchestrator, raised on 2025-09-17. Release Orchestrator finding SEC-2025-02 is remediated, with a closure date of 2025-09-19. SEC-2025-02 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Remediated is the status of finding SEC-2026-17, dated 2026-06-19. Residual risk after remediation of Motion Planner finding SEC-2026-17 is low; the port requires a signed local credential. High is the rating finding SEC-2026-17 carries, at 7.9 on the assessor's own scale. A local compromise could alter planner limits without detection is the exposure SEC-2026-17 describes. finding SEC-2026-17 was one of two items where TDR-2025-38 could not obtain the underlying data within the engagement.

On 2025-09-18 the tester recorded Medium-severity finding SEC-2025-03: remote operations session relay accepted an unpinned certificate chain. In finding SEC-2025-03 the exploitation path means an on-path attacker on a customer network could observe operator commands. The residual position on SEC-2025-03 against Remote Operations Console is low; pinning is applied at the gateway and the agent, stated after mitigation rather than assuming full success. The remediation ledger shows finding SEC-2025-03 as remediated, 2026-01-22. finding SEC-2025-03 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

For Low-severity finding SEC-2026-13 the record shows that throughput degradation measured by the tester, which is moderate evidence and is graded separately from severity. The reason finding SEC-2026-13 was not downgraded is that a large payload degrades broker throughput. Finding SEC-2026-13 is rated Low with a numeric band of 3.3 on the tester's scoring scale. SEC-2026-13 against Order Broker was opened on 2026-04-20 and states that order broker accepted an oversized payload without a limit. finding SEC-2026-13 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Evidence behind Gripper Controller finding SEC-2026-11 is that file permissions observed on a laboratory robot, which the reviewer treats as moderate, because the demonstration was performed in a laboratory tenant rather than in production. Gripper controller parameter store writable by any on-robot process is the substance of finding SEC-2026-11, raised on 2026-04-19. SEC-2026-11 against Gripper Controller translates into the following exposure: a non-privileged process could alter a force limit parameter. The reviewer's position on SEC-2026-11 against Gripper Controller is that the underlying pattern appears in two other components that the tester did not reach. Gripper Controller finding SEC-2026-11 appears twice in the working papers for TDR-2025-38 because the first answer was corrected after a follow-up request.

What remains after Identity and Authorization finding SEC-2026-08 is remediated is low; role grants now require a second approver and are alerted. Reading SEC-2026-08 against Identity and Authorization, the defect is that identity service admin role assignable without a second approval; the discovery date is 2026-04-18. On Identity and Authorization finding SEC-2026-08 the conclusion is that the item is open past its committed date and no revised date has been agreed. SEC-2026-08 arrives as a Medium finding with a 6.4 score, and the reviewer did not adjust either. SEC-2026-08 would move from a concern to a verified defect if one further measurement were supplied.

SEC-2026-07 against Hardware-in-the-Loop Rig is not fully retired by its remediation: medium; network segmentation is the only current control. Medium-severity finding SEC-2026-07 translates into the following exposure: an unpatched laboratory host sits on the engineering network. As at the review date SEC-2026-07 was open and the closure entry reads committed 2026-10-31. Medium-severity finding SEC-2026-07 is supported by the fact that operating system version observed on the rig; evidence strength is strong. Medium-severity finding SEC-2026-07 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

SEC-2025-07 against Site Provisioning Service is remediated, with a closure date of 2026-01-16. The reason SEC-2025-07 against Site Provisioning Service was not downgraded is that the same secret was reusable at any site provisioned from that template. Assessment finding SEC-2025-07 sits against Site Provisioning Service and records that site provisioning stored a shared bootstrap secret in the template. Severity on Site Provisioning Service finding SEC-2025-07 is Medium, scored 5.9 by the assessor. assessment finding SEC-2025-07 illustrates why TDR-2025-38 separates the severity of a finding from the confidence behind it.

Schedule L - Vulnerability ageing against the internal remediation window (TDR-2025-38)

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-01	High	22	30 days	yes	manual review before release
SEC-2025-02	High	142	30 days	no	network segmentation
SEC-2025-03	Medium	87	90 days	yes	audit trail only
SEC-2025-04	Medium	43	90 days	yes	manual review before release
SEC-2025-05	High	76	30 days	no	manual review before release
SEC-2025-06	Medium	22	90 days	yes	second delivery path
SEC-2025-07	Medium	24	90 days	yes	none recorded
SEC-2025-08	Medium	154	90 days	no	network segmentation
SEC-2025-09	Medium	308	90 days	no	network segmentation
SEC-2025-10	Low	167	180 days	yes	monitoring alert on use
SEC-2025-11	Low	78	180 days	yes	manual review before release
SEC-2025-12	Low	354	180 days	no	monitoring alert on use
SEC-2026-01	High	232	30 days	no	monitoring alert on use
SEC-2026-02	High	127	30 days	no	none recorded
SEC-2026-03	High	27	30 days	yes	monitoring alert on use
SEC-2026-04	Medium	135	90 days	no	restricted role membership
SEC-2026-05	Medium	100	90 days	no	second delivery path
SEC-2026-06	Medium	192	90 days	no	manual review before release
SEC-2026-07	Medium	158	90 days	no	monitoring alert on use
SEC-2026-08	Medium	151	90 days	no	monitoring alert on use
SEC-2026-09	Low	150	180 days	yes	restricted role membership
SEC-2026-10	Low	78	180 days	yes	restricted role membership
SEC-2026-11	Low	200	180 days	no	restricted role membership
SEC-2026-12	Low	252	180 days	no	monitoring alert on use
SEC-2026-13	Low	24	180 days	yes	audit trail only
SEC-2026-14	Low	128	180 days	yes	audit trail only
SEC-2026-15	Low	127	180 days	no	none recorded
SEC-2026-16	Medium	148	90 days	no	restricted role membership
SEC-2026-17	High	147	30 days	no	none recorded
SEC-2026-18	Low	305	180 days	no	audit trail only
SEC-2026-19	Medium	116	90 days	no	none recorded
SEC-2026-20	Low	97	180 days	no	monitoring alert on use

4. Authentication, authorization and the trust boundaries

For Release Orchestrator the supportability position is none identified. Release Orchestrator reports 55 per cent line coverage, measured by the same instrumentation the pipeline uses. The manifests place Release Orchestrator cloud, single

global deployment. Interface discipline around Release Orchestrator is adequate; the schema is versioned and change is gated on a compatibility check. What the reviewer had for Release Orchestrator was the release policy, ninety release records and an audit of emergency-path usage; the conclusions below do not reach past it. Release Orchestrator was checked against the company's own register, which had not been updated since the change landed.

Of the 16 open items against Release Orchestrator, the reviewer treats one as material: an emergency release path allows a single approver, and it was used more often than the policy expects. Ownership of Release Orchestrator sits with Developer Productivity; Colin Achebe signs off changes to it. Release Orchestrator depends on the build system, the simulation harness gate and the identity service and is depended on by the OTA updater and the cloud deployment controllers. Developer Productivity releases Release Orchestrator continuous, self-hosting, and no release in the sampled window bypassed the gate set. PostgreSQL 14 for release records and object storage for artefacts is the system of record for Release Orchestrator, a Go 1.22 component running on cloud, a pipeline runner with approval hooks. Release Orchestrator is included because a reader of TDR-2025-38 should be able to disagree with the inference without repeating the work.

Edge Gateway Agent releases every three weeks. The topology for Edge Gateway Agent is edge, one per robot, with mutual authentication to the cloud, and the reviewer confirmed it against the deployment manifests. Reading the dependency map from Edge Gateway Agent outward gives the on-robot services that emit telemetry on the input side and the telemetry ingest path and the robot shadow store on the output side. Coverage for Edge Gateway Agent was 74 per cent line coverage at the review date, and the trend over the reviewed period was up by four points. Where Edge Gateway Agent is concerned, the commitment is that no telemetry is lost for an outage shorter than four hours; the reviewer traced the metric to its source. Edge Gateway Agent was checked against the corresponding entry in the company's own register, and the register was current.

Edge Gateway Agent is owned by the Platform Infrastructure team, with Sam Oyelaran named as the engineering owner of record. For Edge Gateway Agent the reviewer worked from the agent design note, buffer sizing calculations and a four-hour offline test. Read against its objective, Edge Gateway Agent the design is sound and the test evidence is thinner than the design deserves. The interface Edge Gateway Agent publishes is versioned, and the telemetry ingest path and the robot shadow store were checked for use of any undocumented field. Against Edge Gateway Agent the backlog runs to 5 items, and it includes a known behaviour whereby buffer pressure is reported as a robot-level metric and is not aggregated to a site view. Edge Gateway Agent was raised again with the owning engineer after the initial interview, and the position did not change.

The WMS connector and the site configuration service feed Order Broker; a failure in Order Broker is felt by the fleet allocator and the analytics warehouse loader. For Order Broker the reviewer records that the component is adequate today and carries a supportability liability that is not on the funded plan. Order Broker commits that no accepted order is lost, measured by journal reconciliation each night, and the objective is measured continuously rather than sampled. Fleet Services releases Order Broker weekly, and no release in the sampled window bypassed the gate set. The ageing position for Order Broker is none identified, and the reviewer checked it against the vendor statements on file. Order Broker was excluded from an earlier draft of TDR-2025-38 and reinstated after the reviewer obtained the supporting record.

Order Broker was built in Go 1.22 and has not been rewritten; it still runs on cloud, containers on a managed Kubernetes cluster against PostgreSQL 14 with an event journal. Order Broker reports 78 per cent line coverage, measured by the same instrumentation the pipeline uses. Order Broker rests on the broker journal schema, the nightly reconciliation output and two cancellation incidents so far as this review is concerned. 11 defects remain open against Order Broker; the one the reviewer pressed on is that order cancellation races with assignment and can leave an orphaned task record. Grace Lindqvist of Fleet Services maintains Order Broker, and the service inventory entry was current at the review date. Order Broker is the clearest instance in TDR-2025-38 of engineering practice being better than the documentation suggests.

14 defects remain open against Alarm Router; the one the reviewer pressed on is that an on-call rotation gap in the routing table falls back to a shared mailbox nobody watches at night. For Alarm Router the reviewer worked from the routing table export, delivery receipts for one quarter and two missed-page reviews. The paging provider and the customer notification channel rely on Alarm Router, which in turn relies on the fleet health monitor and the on-call rotation source. Alarm Router is written in Python 3.11, executes on cloud, containers plus a scheduled dispatcher, and

keeps its state in PostgreSQL 14 for routes and delivery receipts. Supportability of Alarm Router turns on one fact: the paging provider's older application interface is still in use. Alarm Router was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Ownership of Alarm Router sits with Field Reliability; Wanda Njoku signs off changes to it. Alarm Router reports 51 per cent line coverage, measured by the same instrumentation the pipeline uses. Alarm Router does not share a schema with any consumer; every read by the paging provider and the customer notification channel goes through the published interface. The cadence recorded for Alarm Router is fortnightly, which matches the observed release history. Field Reliability publishes one objective for Alarm Router: a dispatched alarm reaches an on-call responder inside 90 s at p95. Alarm Router appears twice in the working papers for TDR-2025-38 because the first answer was corrected after a follow-up request.

84 per cent line coverage is reported for Identity and Authorization, and the reviewer re-ran the measurement on a clean checkout. Identity and Authorization was tested against the dependency ageing register, which records none identified in the reviewed manifests. Security Engineering releases Identity and Authorization weekly, with a security review on every change, and no release in the sampled window bypassed the gate set. The evidence base for Identity and Authorization was the authorization model document, a token lifetime audit and the secret rotation log. The topology for Identity and Authorization is cloud, global control plane with regional caches, and the reviewer confirmed it against the deployment manifests. Identity and Authorization was reviewed against the position twelve months earlier, and the direction of travel is favourable.

6 items sit in the Identity and Authorization backlog; the entry that bears on the investment case is that service-to-service tokens are issued with a twelve-hour lifetime, longer than the design intent. For Identity and Authorization the stated target is that token issuance inside 200 ms at p95 and no unauthenticated write path, expressed against production traffic rather than a benchmark. Behind Identity and Authorization sits PostgreSQL 14 with a hardware-backed key store; the service itself is Go 1.22 on cloud, containers on a managed Kubernetes cluster. Every cloud service and the remote operations console rely on Identity and Authorization, which in turn relies on the customer directory federations and the secret store. Identity and Authorization is owned by the Security Engineering team, with Sonia Krall named as the engineering owner of record. Identity and Authorization was reviewed against the position twelve months earlier, and nothing measurable has changed.

Traffic Manager commits that corridor deadlock detected and broken inside two seconds, and the objective is measured continuously rather than sampled. The topology for Traffic Manager is cloud, regional, with a degraded on-robot yield behaviour if the link drops, and the reviewer confirmed it against the deployment manifests. Against Traffic Manager the backlog runs to 12 items, and it includes a known behaviour whereby a corridor reservation leaks when a robot drops off the network mid-traverse. Responsibility for Traffic Manager was confirmed with Owen Brantley, whose team, Fleet Services, also holds the on-call duty. The reviewer's judgement on Traffic Manager is that the design is sound and the test evidence is thinner than the design deserves. Traffic Manager is one of the entries where TDR-2025-38 relies on the company's own measurement rather than an independent one.

The cadence recorded for Traffic Manager is weekly, paired with the allocator release, which matches the observed release history. Test coverage on Traffic Manager is 74 per cent line coverage and the measurement basis was confirmed with Owen Brantley. For Traffic Manager the reviewer worked from the reservation protocol specification and a replay of three deadlock events. Data ownership is clear for Traffic Manager: it writes Redis 7 for corridor reservations with a PostgreSQL 14 audit trail and no other service writes there. Upstream of Traffic Manager sit the fleet allocator and the site map service; downstream, the motion planner and the remote operations console consume its output. Traffic Manager was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Schedule B - Deployment topology and failure domain by service (TDR-2025-38)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Motion Planner	edge, one instance per robot with no cloud fallback	every three weeks, promoted ring by ring	3	one robot	Priya Raghunathan
Costmap Fusion	edge, co-scheduled with the planner on the compute module	monthly, batched with perception model updates	4	one robot	Daniel Okafor

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Allocator	cloud, one regional deployment serving all sites in that region	weekly, behind a change approval	3	one region	Marta Kellerman
Traffic Manager	cloud, regional, with a degraded on-robot yield behaviour if the link drops	weekly, paired with the allocator release	2	one region	Owen Brantley
Charge Scheduler	cloud, regional, with a conservative local fallback policy on the robot	fortnightly, without a formal approval gate until March 2026	2	one region	Nadia Semenova
Safety Controller	edge, dual-channel, with cross-checking between channels	twice a year, with full revalidation of the safety case	4	one robot	Bettina Vogel
E-Stop Chain	edge, hardwired, independent of the compute module	changed only by an engineering change order	3	one robot	Theo Bassett
Lidar Driver	edge, one process per sensor	as needed, historically twice a year	6	one robot	Yuki Harada
Localization Engine	edge, with map distribution from the cloud	every three weeks with the planner	3	one robot	Rahul Menon
WMS Connector	cloud, one deployment per customer tenant	every two weeks, per tenant on request	4	one region	Tomas Villareal
Order Broker	cloud, regional	weekly	1	one region	Grace Lindqvist
Telemetry Ingest	cloud, one ingest endpoint per region with an edge buffer on the robot	weekly, with schema changes gated on a compatibility check	6	one robot	Farida Cisse
OTA Updater	split edge and cloud, with staged rings and a per-site pause control	every three weeks, following the autonomy release	4	one robot	Erik Palmgren
Identity and Authorization	cloud, global control plane with regional caches	weekly, with a security review on every change	5	one region	Sonia Krall
Remote Operations Console	cloud, with a thin session relay in each region	twice a week	3	one region	Anjali Sekhon
Simulation Harness	cloud batch, with a local mode for developers	continuous, from the main branch	2	one region	Colin Achebe
Hardware-in-the-Loop Rig	laboratory, two rigs, one of which lags the current sensor configuration	the harness changes with each firmware release	6	one region	Nathan Oduya

4. Authentication, authorization and the trust boundaries - part 2 of 2

Because the 4. Authentication, authorization and the trust boundaries does not fit one page, its remainder is set out below under the same references. The break is editorial.

Definitions given once for the 4. Authentication, authorization and the trust boundaries are relied on again rather than restated, the compilation basis being identical. Pagination alone determines where it falls.

Schedule B - Deployment topology and failure domain by service (TDR-2025-38) (continued, part 2 of 2)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Health Monitor	cloud, regional	fortnightly	4	one region	Wanda Njoku
Battery Management Service	split edge and cloud	pack firmware twice a year, the cloud service monthly	1	one robot	Peter Halvorsen
Gripper Controller	edge, only on the two robot variants fitted with a manipulator	quarterly	3	one robot	Ines Duarte
Map Authoring	cloud, single global	weekly	6	one region	Lucia Ferrante

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Studio	deployment				
Site Provisioning Service	cloud, global	fortnightly	2	one region	Marcus Duplessis
Event Bus	cloud, one cluster per region, three availability zones	upgraded twice a year	4	one region	Erik Palmgren
Config Distribution	split edge and cloud, with a signed bundle format	weekly	6	one robot	Sam Oyelaran
Robot Shadow Store	cloud, regional, sharded by site	weekly	3	one region	Marta Kellerman
Analytics Warehouse Loader	cloud, one loader per region writing to a shared warehouse	weekly	4	one region	Farida Cisse
Billing Usage Meter	cloud, regional	monthly, aligned to the billing cycle	1	one region	Nadia Semenova
Alarm Router	cloud, regional, with a second delivery provider	fortnightly	3	one region	Wanda Njoku
Video Snippet Service	cloud, regional, with an on-robot capture buffer	monthly	2	one region	Anjali Sekhon
Speed and Separation Monitor	edge, cross-checked against the safety controller's own inputs	twice a year, with revalidation	1	one robot	Bettina Vogel
Wheel Odometry Filter	edge	with the firmware, twice a year	6	one robot	Yuki Harada
Payload Scale Interface	edge, only on robots fitted with the weighing deck	as needed	2	one robot	Ines Duarte
Edge Gateway Agent	edge, one per robot, with mutual authentication to the cloud	every three weeks	4	one robot	Sam Oyelaran
Release Orchestrator	cloud, single global deployment	continuous, self-hosting	4	one region	Colin Achebe

5. Supply chain and dependency scanning results

The distribution question for k8s-operator-framework turns on the fact that it is used cloud deployment controllers. Policy compliance for k8s-operator-framework rests on its approval entry, which is approved 2024-09-24. Where k8s-operator-framework came from matters here: public registry, signature verified. k8s-operator-framework was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The reviewer's note on media-transcode is that the build must exclude the GPL-configured components and the reviewer verified the current build flags. Policy compliance for media-transcode rests on its approval entry, which is approved with conditions 2024-08-16. Media-transcode presents a component with open advisories that do not affect the deployed configuration to the engineering team. media-transcode was checked against the company's own register, which had not been updated since the change landed.

Grid-render carries this qualification: the commercial licence removes the copyleft obligation and the reviewer inspected the licence certificate. The terms applying to grid-render are GPL-3.0 with a commercial alternative; the copyleft classification is strong. Grid-render carries the approval entry approved with conditions 2023-07-19; the reviewer looked for a corresponding decision record. grid-render sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Lidar-vendor-sdk appears at OSS-018 in version 4.1.2; the usage recorded is lidar driver. The supply route for lidar-vendor-sdk is supplied by the sensor vendor. Approval for lidar-vendor-sdk reads approved 2023-05-22, and the reviewer checked the approving record. lidar-vendor-sdk was raised again with the owning engineer after the initial interview, and the position did not change.

Linking mode matters for yaml-parse, which is used configuration parsing on the robot. For yaml-parse the obligation is nothing beyond retaining the licence text with the source, and the reviewer tested whether it is met. Yaml-

parse was assessed for support as well as licence risk, and it carries a build-time dependency whose absence would stop the pipeline rather than the product. yaml-parse was checked against the corresponding entry in the company's own register, and the register was current.

Otel-instrument appears at OSS-023 in version 1.24.0; the usage recorded is tracing across cloud services. A build-time dependency whose absence would stop the pipeline rather than the product attaches to otel-instrument independently of its licence terms. Otel-instrument entered the codebase by this route: public registry, signature verified. otel-instrument was excluded from an earlier draft of TDR-2025-38 and reinstated after the reviewer obtained the supporting record.

Against wms-adapter-sdk the reviewer would add that the seat count in the licence is lower than the current engineering headcount using it. The operational obligation attaching to wms-adapter-sdk is to exclude a particular build configuration from the shipped artefact. On provenance, wms-adapter-sdk shows supplied by the warehouse system vendor under a written licence. wms-adapter-sdk was checked against the company's own register, which had not been updated since the change landed.

A single-maintainer dependency with no succession plan attaches to geom2d-fusion independently of its licence terms. The open question on geom2d-fusion is that dynamic linking preserves the weak-copyleft boundary; counsel confirmed the analysis in 2025. Geom2d-fusion carries LGPL-2.1 terms, classified weak for copyleft purposes. geom2d-fusion was excluded from an earlier draft of TDR-2025-38 and reinstated after the reviewer obtained the supporting record.

Rtos-shim was assessed for support as well as licence risk, and it carries an upstream project that is active but changes interface frequently. The open question on rtos-shim is that no obligations beyond attribution. Rtos-shim sits inside Motion Planner and is therefore in the shipped product rather than in tooling. rtos-shim was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The supply route for protobuf-runtime is public registry, signature verified. In Config Distribution the component protobuf-runtime is used message encoding across edge and cloud, which is the fact the licence analysis rests on. The open question on protobuf-runtime is that attribution present in both notice files. protobuf-runtime was checked against the corresponding entry in the company's own register, and the register was current.

Safety-controller-toolchain carries the approval entry approved 2023-02-10; the reviewer looked for a corresponding decision record. Licensing for safety-controller-toolchain is commercial, development use only and the copyleft column reads no. For safety-controller-toolchain the obligation is nothing beyond retaining the licence text with the source, and the reviewer tested whether it is met. safety-controller-toolchain would be a candidate for the first post-close engineering review under any reasonable plan.

Tracing pager-provider-sdk to its source gives supplied by the paging provider. The non-licence risk on pager-provider-sdk is an upstream project with no active maintainer. The older application interface is deprecated by the provider with an end date in 2027 is the reviewer's observation on pager-provider-sdk. pager-provider-sdk is the clearest instance in TDR-2025-38 of engineering practice being better than the documentation suggests.

6. Security incidents and the follow-through on them

Time to mitigate Evergreen Warehousing incident INC-2025-052 was 4 h 10 m, and time to resolve was 16 days. Attention came to scheduling event INC-2025-052 when the site's own shift report, escalated the next morning. Evergreen Warehousing incident INC-2025-052 was caused by a design gap: the scheduler treated a shift calendar entry as immutable once the window had opened. Incident INC-2025-052 would have been shorter or smaller had it not been that the site had changed its shift pattern without notifying the deployment team. The reviewer grades the Charge Scheduler incident INC-2025-052 evidence as moderate on the cause and strong on the delivery position, which was checked in the repository and has not upgraded the finding to compensate. scheduling event INC-2025-052 is the clearest instance in TDR-2025-38 of a control that exists on paper and is weaker in practice.

Charge Scheduler accounts for incident INC-2025-052, a scheduling event. The blast radius of INC-2025-052 was one site and 44 robots. The remediation committed after incident INC-2025-052 was to release charge windows on a shift-end event and reconcile the calendar nightly, owned by Nadia Semenova with a due date of 2025-06-30. The reviewer's reading of Evergreen Warehousing incident INC-2025-052 is that the blast radius was set by a deployment

topology decision made years earlier. Verification of the Charge Scheduler incident INC-2025-052 remediation gives landed on 2025-06-11 with a nightly reconciliation report. Evergreen Warehousing incident INC-2025-052 was excluded from an earlier draft of TDR-2025-38 and reinstated after the reviewer obtained the supporting record.

Taken with the rest of the record, environmental event INC-2026-068 indicates that the incident is unremarkable in itself and revealing about the observability of the surrounding system. Wheel Odometry Filter incident INC-2026-068 would have been shorter or smaller had it not been that the site's cleaning schedule changed and now overlaps the peak shift. In taxonomy terms environmental incident INC-2026-068 is a environmental incident in Wheel Odometry Filter. Duration for environmental event INC-2026-068 splits into not a service outage; 3 h of degraded operation of impact and open at the review date to close the underlying cause. Verification of the Harborline Systems incident INC-2026-068 remediation gives not landed; the detector is prototyped and unvalidated. INC-2026-068 illustrates why TDR-2025-38 separates the severity of a finding from the confidence behind it.

So far as reach is concerned, incident INC-2026-068 involved one site and 14 robots. The corrective action for Wheel Odometry Filter incident INC-2026-068 is to add a slip detector that flags the condition and issue a site advisory on cleaning windows, and 2026-09-15 is the date the team gave. What went wrong in INC-2026-068 was that the filter has no wet-floor slip model and does not flag the condition to the operator. Detection came through robots stopped and requested assistance, which the site reported as unplanned downtime, which places environmental incident INC-2026-068 on the reported side of the ledger. Wheel Odometry Filter incident INC-2026-068 rests on limited, being a single account corroborated only by the ticket timeline evidence, which the reviewer states separately from the severity of the finding. INC-2026-068 would move from a concern to a verified defect if one further measurement were supplied.

Process bypass incident INC-2025-131 produced one commitment: to restrict export to published revisions and shorten the publication path so the workaround is unnecessary, owned by Lucia Ferrante, dated 2025-12-15. Reading process bypass incident INC-2025-131 from its first entry, on 2025-10-14 an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. Time to mitigate process bypass event INC-2025-131 was 3 h 22 m, and time to resolve was 44 days. For Map Authoring Studio incident INC-2025-131 the reviewer holds strong, since the reviewer reproduced the failure mode in a laboratory environment evidence and reports the finding at that strength. Process bypass event INC-2025-131 affected one site and 34 robots. process bypass event INC-2025-131 is reported here in full because a summary of it in an earlier draft of TDR-2025-38 was misread as an assurance.

Underneath Map Authoring Studio incident INC-2025-131 lies the design decision that the export function did not check publication state. INC-2025-131 carries a second-order finding: the site had a standing workaround for slow map publication that predated the review step. Process bypass incident INC-2025-131 supports one conclusion and not another: the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Reported detection applies to INC-2025-131, since a field engineer reported unexpected keep-out zones. The reviewer checked the process bypass incident INC-2025-131 commitment against the code and release records; the export restriction landed on 2025-11-27; the publication work is open. incident INC-2025-131 was excluded from an earlier draft of TDR-2025-38 and reinstated after the reviewer obtained the supporting record.

Incident INC-2025-152 began on 2025-12-04 when one region retained operator video clips for ninety days against a thirty-day commitment. Scope of impact in data retention event INC-2025-152 was no production site and no robot directly, the exposure being to internal systems, taken from the fleet records rather than from the narrative. In INC-2025-152, bucket creation was not covered by the configuration baseline check, which is a process finding rather than a code finding. Taken with the rest of the record, Video Snippet Service incident INC-2025-152 indicates that an operational workaround masked the defect for longer than the incident record suggests. Data retention is the class of INC-2025-152; Video Snippet Service is the component. incident INC-2025-152 was raised again with the owning engineer after the initial interview, and the position did not change.

Cause analysis in internal incident INC-2025-152 concluded that the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. Grading INC-2025-152 on evidence rather than on consequence gives strong, since the reviewer reproduced the failure mode in a laboratory environment. Time to mitigate internal incident INC-2025-152 was not applicable; retention only, and time to resolve was 37 days. For INC-2025-152 the follow-through was tested rather than assumed, and the rule landed on 2026-01-10; deletion of the historical clips is open. The corrective action for incident INC-2025-152 is to apply the lifecycle rule through the configuration baseline and

delete the over-retained clips, and 2026-01-31 is the date the team gave. data retention incident INC-2025-152 would be a candidate for the first post-close engineering review under any reasonable plan.

WMS Connector incident INC-2025-058 carries strong on the cause and limited on the contributing factor evidence, and the reviewer declined to raise the finding beyond what that supports. WMS Connector incident INC-2025-058 took 2 h 35 m to stabilise and 21 days to close out. INC-2025-058 would have been shorter or smaller had it not been that the integration specification allowed the identifier to be tenant-scoped but did not require it. For incident INC-2025-058 the follow-through was tested rather than assumed, and landed on 2025-05-30; the backfill was completed on 2025-06-03. The post-mortem for Dunbar Industrial incident INC-2025-058 identifies the cause as follows: duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. incident INC-2025-058 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Data integrity event INC-2025-058 is not read as an isolated failure: the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. On 2025-05-09 the record for Dunbar Industrial incident INC-2025-058 opens: two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. Classifying data integrity event INC-2025-058 as data integrity matters because the WMS Connector backlog contains two further items of the same kind. Attention came to incident INC-2025-058 when the customer reported missing picks the same afternoon. Remediation for INC-2025-058 is to key suppression on the tenant and identifier pair and backfill the affected order records; Tomas Villareal owns it and the committed date is 2025-07-15. data integrity incident INC-2025-058 illustrates why TDR-2025-38 separates the severity of a finding from the confidence behind it.

After Analytics Warehouse Loader incident INC-2025-110, Farida Cisse took the action to fail the load on an incomplete partition and reprocess the affected days by 2025-10-31. The contributing factor recorded against data integrity event INC-2025-110 is that the partition completeness check was written but never wired into the load gate. In taxonomy terms data integrity event INC-2025-110 is a data integrity incident in Analytics Warehouse Loader. Analytics Warehouse Loader incident INC-2025-110 was mitigated in not applicable; reporting only; the resolution ran to 18 days. Data integrity event INC-2025-110 rests on moderate, resting on the team's own post-mortem and one supporting log extract evidence, which the reviewer states separately from the severity of the finding. data integrity incident INC-2025-110 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Reading data integrity incident INC-2025-110 from its first entry, on 2025-09-03 a late telemetry partition was excluded from the daily load and a customer report understated throughput. Detection came through the customer challenged the report three days later, which places incident INC-2025-110 on the reported side of the ledger. On delivery, Analytics Warehouse Loader incident INC-2025-110 shows this: landed on 2025-09-21 with a reprocessing run for the affected week. INC-2025-110 touched 1 site or tenant with no robot taken out of service, and Kestrel Medical Supply was the affected account. Cause analysis in Analytics Warehouse Loader incident INC-2025-110 concluded that the loader treated a missing partition as empty rather than as incomplete. data integrity incident INC-2025-110 would move from a concern to a verified defect if one further measurement were supplied.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2025-38)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
give every corridor reservation a lease with an expiry and add a site-level queue depth alert	INC-2025-041	Owen Brantley	2025-05-30	landed	runbook change inspected
re-derive the time offset on every reconnect and add the affected variant to the second bench ri	INC-2025-047	Yuki Harada	2025-06-15	partly landed	runbook change inspected
release charge windows on a shift-end event and reconcile the calendar nightly	INC-2025-052	Nadia Semenova	2025-06-30	landed	configuration export inspected
key suppression on the tenant and identifier pair and backfill the affected	INC-2025-058	Tomas Villareal	2025-07-15	landed	commit inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
order records					
make the allocator consult charger availability before excluding a robot	INC-2025-063	Marta Kellerman	2025-07-31	landed	release record inspected
add a leadership and under-replicated partition check to the broker upgrade runbook	INC-2025-071	Erik Palmgren	2025-07-31	landed	configuration export inspected
invalidate the roster cache on a fleet membership event and add a session-level regression test	INC-2025-078	Anjali Sekhon	2025-08-31	landed	configuration export inspected
persist the pause state and require a design note for any change to the release state machine	INC-2025-084	Erik Palmgren	2025-08-15	landed	interview only
add the connector torque check to incoming inspection and retrofit the affected harness batch	INC-2025-090	Bettina Vogel	2025-09-30	partly landed	no evidence located
replace the mailbox fallback with a secondary responder and import rotations from the scheduling	INC-2025-096	Wanda Njoku	2025-10-15	partly landed	no evidence located
add a deterministic tie-break and return the pinch-point scenario to the nightly suite	INC-2025-103	Priya Raghunathan	2025-10-31	partly landed	test added and observed
fail the load on an incomplete partition and reprocess the affected days	INC-2025-110	Farida Cisse	2025-10-31	landed	runbook change inspected
redact authorization headers at the logging layer and cut build log retention to fourteen days	INC-2025-118	Sonia Krall	2025-10-31	partly landed	configuration export inspected
annotate staleness on every read path and add a rebalance-under-load test to the platform suite	INC-2025-124	Marta Kellerman	2025-11-30	landed	commit inspected
restrict export to published revisions and shorten the publication path so the workaround is unn	INC-2025-131	Lucia Ferrante	2025-12-15	partly landed	release record inspected
split control traffic onto a dedicated topic with its own quota and revisit the topic inventory	INC-2025-139	Erik Palmgren	2026-01-31	partly landed	interview only
fold the reset into the re-run command and make double-counting detectable in the ledger	INC-2025-146	Nadia Semenova	2025-12-31	landed	commit inspected
apply the lifecycle rule through the configuration baseline and delete the over-retained	INC-2025-152	Anjali Sekhon	2026-01-31	partly landed	runbook change inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
clips					

6. Security incidents and the follow-through on them - part 2 of 2

A further extract of the 6. Security incidents and the follow-through on them appears here, taken in order from the point at which the previous part ended. The break is editorial.

Scope, cut-off and units for the 6. Security incidents and the follow-through on them were fixed at its first part and are not repeated, being unaltered. The break is editorial.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2025-38) (continued, part 2 of 2)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
replace the fallback matcher and re-survey the affected aisles	INC-2026-004	Rahul Menon	2026-06-30	not landed	test added and observed
require a calibration record before returning a manipulator robot to service	INC-2026-009	Ines Duarte	2026-04-15	landed	configuration export inspected
move precondition validation to apply time and add an offline apply case to the platform suite	INC-2026-015	Sam Oyelaran	2026-03-31	landed	commit inspected
pin the full build configuration and add a determinism check to the harness	INC-2026-021	Colin Achebe	2026-05-31	not landed	interview only
group on the signal and block closure while a constituent alert remains active	INC-2026-028	Wanda Njoku	2026-04-30	landed	configuration export inspected
measure closing rate where the sensing permits and re-run the protective distance calculation	INC-2026-033	Bettina Vogel	2026-08-31	not landed	configuration export inspected
serialise cancellation against assignment and add an hourly orphan sweep	INC-2026-039	Grace Lindqvist	2026-05-31	landed	release record inspected
aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current h	INC-2026-045	Sam Oyelaran	2026-06-30	partly landed	configuration export inspected
check every template against the baseline and re-converge the four forked templates	INC-2026-051	Marcus Duplessis	2026-06-15	partly landed	test added and observed
refit the estimator on opportunity-charge data and add an imbalance alert independent of state o	INC-2026-057	Peter Halvorsen	2026-09-30	not landed	runbook change inspected
require a justification and a second approver within one working day for every emergency release	INC-2026-062	Colin Achebe	2026-08-31	not landed	commit inspected
add a slip detector that flags the condition and issue a site advisory on cleaning windows	INC-2026-068	Yuki Harada	2026-09-15	not landed	release record inspected
Partition the fleet allocator write path	TD-01	engineering leadership	2027-03-31	in design	register entry only

Commitment	Source	Owner	Committed date	Status	Evidence inspected
Replace the unmaintained scan-matching library in the localization fallback	TD-02	engineering leadership	2026-12-31	committed	register entry only
Retire the vendored geometry library in costmap fusion	TD-03	engineering leadership	2027-06-30	backlog	register entry only
Raise robot software line coverage above sixty per cent on the perception path	TD-04	engineering leadership	2027-03-31	in progress	register entry only
Bring the second bench rig to the current sensor configuration	TD-05	engineering leadership	2026-09-30	committed	register entry only
Make the nightly simulation suite blocking	TD-06	engineering leadership	2026-11-30	blocked	register entry only

7. Security debt and the funded remediation plan

The plan treats Alarm Router item TD-15 as unfunded and dependent on capital rather than engineering effort, which the reviewer checked against the headcount model. Committed is the status of security item TD-15, targeted at 2026-09-30. The reviewer's qualification on Alarm Router item TD-15 is that raised in the 2026 assessment and accepted for remediation. security item TD-15 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Costmap Fusion item TD-04 carries an estimate of 2 engineer-quarters; the costed figure is $240,000. Under quality, entry quality item TD-04 is to raise robot software line coverage above sixty per cent on the perception path. In progress is the status of programme item TD-04, targeted at 2027-03-31. register item TD-04 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Under reliability, entry register item TD-23 is to split the shared database instance used by the allocator and the charge scheduler. Reliability item TD-23 is behind a supplier release outside the company's control, which is why the target date is later than the effort estimate implies. The reviewer's comment on TD-23 is that raised by the reviewer rather than present on the company's own register. Charge Scheduler item TD-23 carries a follow-up request that was satisfied after the draft of TDR-2025-38 was circulated.

On funding, programme item TD-10 is unfunded and dependent on capital rather than engineering effort. Analytics Warehouse Loader item TD-10 carries an estimate of 0.5 engineer-quarters; the costed figure is $54,000. Committed is the status of TD-10, targeted at 2026-12-31. register item TD-10 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Battery Management Service item TD-21 is partly funded, with the second phase unbudgeted in the engineering plan the reviewer was shown. Battery Management Service item TD-21 is in progress with a target date of 2026-09-30. Battery Management Service item TD-21 covers the work to refit the battery state-of-health estimator on opportunity-charge data. programme item TD-21 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Reading the programme plan, register item TD-19 is behind a platform change that has its own target date. Reading Speed and Separation Monitor item TD-19 against the funded plan, the most significant open safety item in the reviewed record. Speed and Separation Monitor carries register item TD-19, the work to measure closing rate for the separation monitor. TD-19 on Speed and Separation Monitor was raised again with the owning engineer after the initial interview, and the position did not change.

Item programme item TD-20 in the debt register is to validate and ship the wheel slip detector, and it bears on Wheel Odometry Filter. The plan treats TD-20 as funded in the current financial year, which the reviewer checked against the headcount model. Sequencing of TD-20 matters: it is behind a platform change that has its own target date. programme item TD-20 was raised again with the owning engineer after the initial interview, and the position did not change.

Whether TD-06 happens depends on funding, and it is not funded, and carried in the register regardless. As at the review date Simulation Harness item TD-06 was blocked, with 2026-11-30 still the stated target. Programme item TD-06 is behind a supplier release outside the company's control, which is why the target date is later than the effort estimate implies. Simulation Harness item TD-06 was reviewed against the position twelve months earlier, and nothing measurable has changed.

8. Findings management accepted or declined to fix

The remediation ledger shows Medium-severity finding SEC-2025-09 as partially remediated, pending. Assessment finding SEC-2025-09 rests on the observation that key custody was evidenced by interview and the key ceremony record; the reviewer holds that to be moderate. Weighing SEC-2025-09 against OTA Updater against the compensating controls, the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. Rating for SEC-2025-09 against OTA Updater is Medium, band 6.3; the reviewer re-read the scoring rationale and accepted it. OTA Updater finding SEC-2025-09 is reported here in full because a summary of it in an earlier draft of TDR-2025-38 was misread as an assurance.

Battery pack diagnostic interface has no rate limit is the substance of Battery Management Service finding SEC-2026-18, raised on 2026-04-23. Battery Management Service finding SEC-2026-18 produces this judgement: management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Battery Management Service finding SEC-2026-18 rests on the observation that flood produced on a bench pack; the reviewer holds that to be strong. The tester scored SEC-2026-18 at 3.4 and labelled it Low. SEC-2026-18 is the clearest instance in TDR-2025-38 of a control that exists on paper and is weaker in practice.

After the fix, finding SEC-2026-16 carries residual risk described as low; the socket binds to the loopback interface only. For Costmap Fusion finding SEC-2026-16 the reviewer notes that the severity is right and the ownership is unclear, since two teams each believe the other holds it. Impact of SEC-2026-16 against Costmap Fusion is not theoretical; on-network access to a live perception diagnostic stream. At 5.8 the score on SEC-2026-16 places it in the Medium group. finding SEC-2026-16 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The remediation ledger shows Traffic Manager finding SEC-2026-14 as remediated, 2026-06-18. After the fix, Traffic Manager finding SEC-2026-14 carries residual risk described as low. SEC-2026-14 records that traffic manager admin action lacked an audit record, discovered on 2026-04-21. Rating for finding SEC-2026-14 is Low, band 3.8; the reviewer re-read the scoring rationale and accepted it. Low-severity finding SEC-2026-14 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

What remains after Site Provisioning Service finding SEC-2026-15 is remediated is low; management judged the operational cost of enforcement higher than the risk. Finding SEC-2026-15 matters because operator confusion rather than a direct security impact. Finding SEC-2026-15 is rated Low with a numeric band of 3.0 on the tester's scoring scale. For finding SEC-2026-15 the record shows that collision created in a staging tenant, which is strong evidence and is graded separately from severity. assessment finding SEC-2026-15 is the entry that changed most between the first and second document requests behind TDR-2025-38.

The practical consequence in finding SEC-2025-05 is that a forged agent could inject telemetry attributed to a real robot. Read alongside the remediation record, SEC-2025-05 shows that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. 2026-02-06 is the closure position on assessment finding SEC-2025-05, which the ledger records as remediated. Moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence supports SEC-2025-05 against Telemetry Ingest, namely that the tester submitted synthetic telemetry from an unenrolled host. Telemetry Ingest finding SEC-2025-05 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The tester scored assessment finding SEC-2026-01 at 7.1 and labelled it High. The reviewer declines to soften High-severity finding SEC-2026-01: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. SEC-2026-01 is not fully retired by its remediation: high until the second-approver control lands. The evidence for finding SEC-2026-01 is strong: the tester used the path in a staging tenant and the audit trail recorded no

justification. SEC-2026-01 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-38.

On SEC-2026-19 against Localization Engine the conclusion is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. SEC-2026-19 against Localization Engine sits against Localization Engine and records that localization map cache stored unencrypted on the robot. The remediation ledger shows SEC-2026-19 as open, committed 2026-10-15. SEC-2026-19 arrives as a Medium finding with a 5.1 score, and the reviewer did not adjust either. assessment finding SEC-2026-19 would move from a concern to a verified defect if one further measurement were supplied.

9. Where a security failure would become a safety failure

Tolerable with the measured approach speed assumption stated is the residual position on hazard HZ-01 in the register, whose status is verified. One caveat attaches to register entry HZ-01: the fixed approach speed assumption is carried as an open qualification. On the register's own scale, Frequent-exposure hazard HZ-01 is Serious injury severity, Frequent exposure, Low controllability. register entry HZ-01 was cross-checked against the engineering record supplied for TDR-2025-38, and the two agree.

The six sites with narrow-aisle geometry is the exposed population for HZ-06. Hazard HZ-06 in the register carries an open qualification: one site had two lockout devices for eleven robots at the time of the visit. Protection at hazard HZ-06 comes from maintenance mode with a physical lockout and a two-step exit confirmation. register entry HZ-06 rests on a single source, and the reviewer has said so rather than implying corroboration.

Equipment damage harm, Occasional exposure and Medium controllability are the parameters recorded for register entry HZ-07. Where HZ-07 is concerned, the hazard is that robot travels into an unmapped area and contacts fixed plant. On history, HZ-07 produced a single event before the current control was fitted. register entry HZ-07 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-38.

Mitigation for Frequent-exposure hazard HZ-08 is corridor reservation with deadlock detection and a deterministic yield tie-break. Risk parameters for HZ-08 are Equipment damage severity, Frequent exposure and High controllability. The residual entry for register entry HZ-08 reads Tolerable, verified. hazard HZ-08 in the register would move from a concern to a verified defect if one further measurement were supplied.

Status partially verified and residual risk Tolerable where cleaning compliance is evidenced are what the register gives for hazard HZ-20. HZ-20 is assessed at Serious injury severity with Occasional exposure and Medium controllability. The record shows that hazard HZ-20 produced one event whose investigation is still open. Occasional-exposure hazard HZ-20 has no single accountable owner, and that is the finding the reviewer would press hardest.

Force bench test after every calibration, evidenced by a calibration record is what verified HZ-04. Serious injury hazard HZ-04 is mitigated by force limiting with a mechanical release and an interlock on the access panel. The parameter set for register entry HZ-04 reads Serious injury, Remote, Low. hazard HZ-04 in the register is included because a reader of TDR-2025-38 should be able to disagree with the inference without repeating the work.

Verification of Improbable-exposure hazard HZ-10 was by supplier test report plus an internal witnessed disconnect test. Fire hazard HZ-10 depends for its mitigation on contactor sequencing with an arc-suppression design in the charge dock. In the reviewed period hazard HZ-10 in the register produced a single event before the current control was fitted. Improbable-exposure hazard HZ-10 is the entry that changed most between the first and second document requests behind TDR-2025-38.

Register entry HZ-11 applies to the six sites with narrow-aisle geometry. In the reviewed period Improbable-exposure hazard HZ-11 produced one field event that did not result in harm. Hazard HZ-11 depends for its mitigation on cell temperature limits, pack-level current derating and a dock thermal cutout. Improbable-exposure hazard HZ-11 is one of the entries where TDR-2025-38 relies on the company's own measurement rather than an independent one.

In the reviewed period HZ-03 produced no event, and the reviewer regards the absence as weak evidence given the exposure period. The scope of Serious injury hazard HZ-03 is every robot variant currently in production. Verification of register entry HZ-03 was by bench drop testing plus a payload limit audit across the fitted fleet. Occasional-exposure

hazard HZ-03 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

10. Open findings, risk acceptances and qualifications

Report TDR-2025-38 closes with 9 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. 9 matters remain open at the date of this report, and none of them is closed by the passage of time alone. One limitation of report TDR-2025-38 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. Where TDR-2025-38 reports a supplier's test result, the reviewer read the certificate and did not witness the test. K. Vandermeer has avoided aggregating findings into a single score, because the aggregate would conceal the distribution.

Not every finding in this report is accepted by management: 2 are declined, and the declinatures are reproduced rather than argued away. TDR-2025-38 records 2 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. This report proposes 7 metrics for the post-close monitoring pack, each already produced by an existing system. Reporting under this report should distinguish a target that was met from a target that was lowered, which the current pack does not. Brandywine Technology Partners was given read access without restriction on scope within the agreed subject matter.

The present report reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. Grading in this report is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. K. Vandermeer tested a sample rather than the population in TDR-2025-38, and the sampling basis is stated in each section. The working method behind TDR-2025-38 was to start from the incident record and work back to the design, rather than the other way round. K. Vandermeer interviewed 11 people for the report at TDR-2025-38; where an account was uncorroborated it is marked as such.

K. Vandermeer would add one qualitative item to the pack proposed in the report at TDR-2025-38: the count of commitments closed without verifying evidence. 7 indicators are recommended in report TDR-2025-38; the reviewer deliberately avoided proposing anything that needs a new pipeline. Brandywine Technology Partners concludes, on the material examined, that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. For Ardent Peak's purposes, TDR-2025-38 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The present report deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

11. Conclusions on the security position

Stated without qualification for once, the conclusion of report TDR-2025-38 is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. TDR-2025-38 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. 4 sites contributed evidence to the report at TDR-2025-38, and one further site was in acceptance testing and was excluded. Sampling in report TDR-2025-38 covered 14 of the 34 catalogued services and 4 of the 18 live sites. The reader of TDR-2025-38 should treat each conclusion as bounded by the section it appears in.

The costed remediation implied by this report is $1,005,000 over the coming four quarters, which is within the ordinary engineering budget. Brandywine Technology Partners notes that 1 of the matters in the present report are capable of being closed before completion if the company is asked in time. Evidence gathering for the report at TDR-2025-38 was iterative: an initial document request, 11 interviews, then a second request that closed most of the gaps. This report did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. K. Vandermeer reports management's own position wherever it differs from the reviewer's, and does not merge the two.

Post-close monitoring under the report at TDR-2025-38 would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. The pack the present report proposes is deliberately short, because a longer one would be produced and not read. Brandywine Technology Partners delivered report TDR-2025-38 on 2026-01-23; the scope agreed with Ardent Peak was the security architecture, penetration-test findings and vulnerability management of the platform. The report at TDR-2025-38 addresses the security architecture,

penetration-test findings and vulnerability management of the platform. It does not address commercial terms, financial reporting or employment matters, which sit with other workstreams. K. Vandermeer has resisted rounding the findings towards a comfortable conclusion in either direction.

The reviewer's judgement on Charge Scheduler is that the component would not block the transaction, and it would appear on any competent post-close engineering plan. The topology for Charge Scheduler is cloud, regional, with a conservative local fallback policy on the robot, and the reviewer confirmed it against the deployment manifests. 7 items sit in the Charge Scheduler backlog; the entry that bears on the investment case is that opportunity-charge windows are not released when a shift ends earlier than planned. 63 per cent line coverage is reported for Charge Scheduler, and the reviewer re-ran the measurement on a clean checkout. Charge Scheduler releases fortnightly, without a formal approval gate until March 2026. Charge Scheduler was cross-checked against the engineering record supplied for TDR-2025-38, and the two agree.

For Lidar Driver the reviewer records that the reviewer accepts the engineering position while noting that the verifying test does not exist. Placement of Lidar Driver is edge, one process per sensor, which the reviewer tested by tracing a request end to end. Ownership of Lidar Driver sits with Robot Firmware; Yuki Harada signs off changes to it. The sensor hardware and the time synchronisation daemon feed Lidar Driver; a failure in Lidar Driver is felt by the costmap fusion pipeline and the localization engine. The cadence recorded for Lidar Driver is as needed, historically twice a year, which matches the observed release history. Lidar Driver would move from a concern to a verified defect if one further measurement were supplied.